



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: PROJECT EVALUATION (24CS2235F)

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TITLE OF THE PROJECT	Mini Compiler for a Custom Programming Language	
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Abstract

The Mini Compiler for a Custom Programming Language is a compiler design project that implements the major phases of a compiler for a simple, user-defined programming language. The system takes source code written in the custom language and translates it into optimized target code through a sequence of compilation phases.

The project begins with lexical analysis, where the source program is converted into tokens using regular expressions and finite automata (DFA). These tokens are then processed by a parser based on a context-free grammar to verify the syntactic structure of the program and generate an Abstract Syntax Tree (AST). The AST is used as an intermediate representation for translating the program into Three Address Code (TAC).

The generated intermediate code is then optimized using techniques such as constant folding, constant propagation, common subexpression elimination, and dead code elimination. Finally, the optimized intermediate code is converted into simple target/machine-like code through code generation and basic register allocation.

The proposed compiler demonstrates the complete compilation pipeline, from source code to target code, and provides practical implementation of important Compiler Design concepts, including lexical analysis, finite automata, parsing, syntax-directed translation, intermediate code generation, code optimization, and code generation. The project aims to provide a clear and practical understanding of how modern compilers transform high-level programs into executable low-level representations.

Signature of the Guide

HOD